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6

Tabu Search

Summary

This chapter describes the tabu search (TS), which is a meta-heuristic algorithm for combinatorial optimization. A brief literature review of development and applications of the TS opens this chapter. The foundations of the TS and its algorithmic steps are described next, followed by a pseudocode of the TS.

6.1 Introduction

The tabu search (TS) was developed by Glover (1986). It is designed to solve combinatorial (finite solution set) optimization problems. Simple TS and advanced TS were introduced by Glover (1989, 1990). Bland (1995) implemented the TS in a structural design context and showed that the TS is a technically viable technique for optimal structural design. Fanni et al. (1999) applied the TS coupled with deterministic strategies for the optimal design of magnetic resonance imaging (MRI) devices. Wang et al. (1999) demonstrated the capability of the TS to optimal design of multiproduct batch chemical processes. Nara et al. (2001) applied the TS to determine locations and discrete capacities of distributed generators so that the distribution loss is minimized. Hajji et al. (2004) developed a new TS algorithm for global optimization of multimodal functions with continuous domain. Misevicius (2005) implemented the TS to solve the quadratic assignment problem (QAP). Nourzad and Afshar (2009) proposed a probabilistic improvement to the neighborhood selection of the TS and used that to find optimal water resources allocation for an industrial copper complex distribution system. Hajji et al. (2010) designed software based on scatter search, the TS, and neural networks for determining water-pumping schedules. Martinez et al. (2010) employed the TS algorithm to optimize water level monitoring stations in lakes and streams within the south Florida water

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management district. Yeh et al. (2013) used the TS and the SA to the optimization of sewer-network designs that are complex nonlinear problems. Haghighi and Bakhshipour (2014) presented an integrated optimization model for designing sewage collection networks and applied the TS method as a deterministic combinatorial meta-heuristic to solve the optimization model.

6.2 Tabu Search (TS) Foundation

The TS is an enhancement of local search (LS) methods. The LS refers to iterative procedures that start with a solution and then employs local modifications (moves) to find a superior solution. The basic concept of the LS is that the movement is always from a worse solution to a better one. The search terminates when it reaches an optimum with respect to the movements made. The optimum achieved with the LS is mostly a local optimum instead of a global optimum given that the algorithm always moves to an improved neighboring solution near the current one. The TS solved the problem of convergence to local optima experienced with LS methods by allowing movements to non-improving solutions when there is no better solution near the current solution. The TS also takes advantage of principles of artificial intelligence by making search movements based on memory structures that prevent repetitive movements and help to explore the decision space of the optimization problem more thoroughly. Previously visited solutions are known as tabu and moving back to them is prevented by the memories that save the history of the search for optima.

TS designates solutions of an optimization problem as points in an N-dimensional space where N denotes the number of decision variables. Neighboring points refer to new solutions. The process of going from the searching point, which is the old solution, to a neighboring point is called a move. The best point reached in the search is the best solution found during the search. Table 6.1 lists the characteristics of the TS.

A simple TS starts by generating a random solution (initial solution), which is known as the searching point. The current searching point is considered momentarily as the best point. In the next step neighboring points are generated near the searching point. Only the neighboring points that are not tabu are considered. The searching point moves to the best neighboring point that is not tabu. Unlike the LS, in the TS the best neighboring point replaces the searching point even if it is worse than the current searching point. The previous searching point is memorized as tabu. If the new searching point is better than the best point, the new point replaces the best point; otherwise the best point remains unchanged. Neighboring points near the new searching point are generated. The previously mentioned process repeats until the termination criteria are satisfied. Figure 6.1 illustrates the flowchart of the TS.

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Table 6.1 The characteristics of the TS.

General algorithm (see Section 2.13)	Tabu search
Decision variable	Point's position
Solution	Point
Old solution	Searching point
New solution	Neighbor point
Best solution	Best point
Fitness function	Desirability of the point
Initial solution	Random point
Selection	Tabu list
Process of generating new solution	Movement

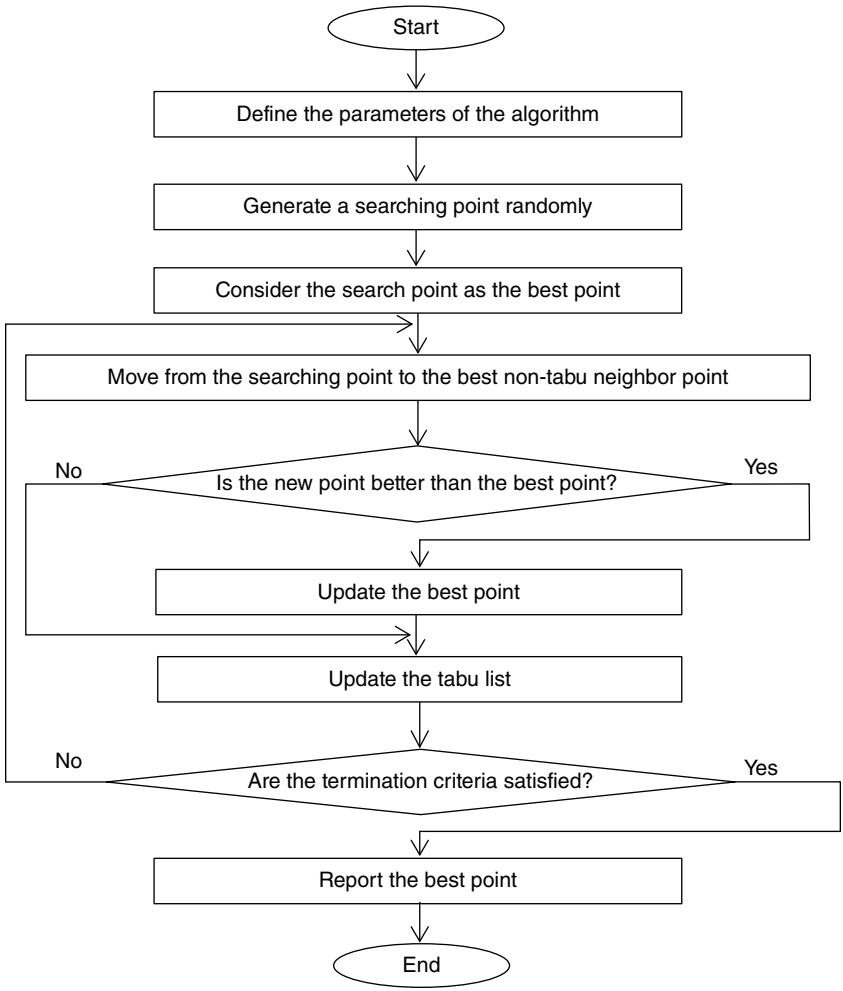


Figure 6.1 The flowchart of the TS.

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6.3 Generating an Initial Searching Point

Each possible solution of the TS is a point in the feasible space of the optimization problem. In an N-dimensional optimization problem, a solution point is an array of size $1 \times N$. The TS starts with a single solution of size $1 \times N$:

$$Point = X = \left(x_1, x_2, \ldots, x_i, \ldots, x_N\right) \tag{6.1}$$

where X = a solution of the optimization problem, $x_i = i$th decision variable of the solution X, and N = number of decision variables. The decision variable values $(x_1, x_2, x_3, \ldots, x_N)$ are represented as a predefined set of values for discrete problems. The optimization algorithm starts with an initial point known as the searching point that is randomly selected from the discrete decision space when solving combinatorial problems (see Section 2.6). Neighboring points (new solutions) are considered near the searching point.

6.4 Neighboring Points

The TS was developed to solve combinatorial (finite solution set) optimization problems. The decision variables take discrete values in this instance. The neighborhood of a solution is made of all the solutions in which the value of one decision variable is changed to its immediate adjacent values in a sorted list of discrete values. Assume that Figure 6.2 portrays the decision space of an optimization problem with a two-dimensional discrete decision space. In Figure 6.2 the circles denote possible answers for the problem.

Suppose that in Figure 6.2, the solution (3,2), which is encircled by a square, is selected as searching point. All the possible solutions that are connected

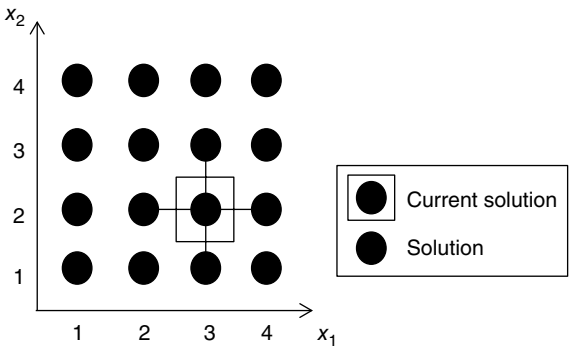


Figure 6.2 The decision space of a two-dimensional optimization problem.

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with lines to (3,2) are neighboring solutions. In an N-dimensional space if $X=(x_1,x_2,\ldots,x_i,\ldots,x_N)$ is the searching point, the number of neighbor solutions surrounding the searching point is $2N$, which are represented as follows:

$$M=\begin{bmatrix} X_1^{(new)}=(x_1',x_2,\ldots,x_i,\ldots,x_N) \\ X_2^{(new)}=(x_1,x_2',\ldots,x_i,\ldots,x_N) \\ \vdots \\ X_i^{(new)}=(x_1,x_2,\ldots,x_i',\ldots,x_N) \\ \vdots \\ X_N^{(new)}=(x_1,x_2,\ldots,x_i,\ldots,x_N') \\ X_{N+1}^{(new)}=(x_1'',x_2,\ldots,x_i,\ldots,x_N) \\ X_{N+2}^{(new)}=(x_1,x_2'',\ldots,x_i,\ldots,x_N) \\ \vdots \\ X_{N+i}^{(new)}=(x_1,x_2,\ldots,x_i'',\ldots,x_N) \\ \vdots \\ X_{2N}^{(new)}=(x_1,x_2,\ldots,x_i,\ldots,x_N'') \end{bmatrix} \tag{6.2}$$

$$x_i'=x_i+\varepsilon,\quad i=1,2,\ldots,N \tag{6.3}$$

$$x_i''=x_i-\varepsilon,\quad i=1,2,\ldots,N \tag{6.4}$$

in which M = the matrix of the neighbor points around X, $X_i^{(new)}=i$th neighboring solution near the searching point X, x_i = the value of decision variable ith of the searching point, x_i' and x_i'' = the new values of decision variable ith, and ε = the length of steps in a discrete decision space.

The search movements are generalized as follows:

$$X^{(new)}=X\pm(\varepsilon\cdot e_{1\times N}) \tag{6.5}$$

in which $X^{(new)}$ = a neighbor solution around X, e = a specified direction vector that is a matrix of zeroes and ones, and ε = the length of steps in a discrete decision space.

For specific combinatorial problem at hand, various alternative possible neighborhood structures can be used according to the search space. Choosing a search space and a neighborhood structure has a key role in the design of any TS algorithm.

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6.5 Tabu Lists

The TS features principles of artificial intelligence because it acts based on memory structures that prevent repetitive movements, thus improving the search for optima within the decision space of the optimization problem more effectively. The TS prevents moving back to previously visited solutions using memories recorded in a tabu list that memorize the history of the search. The tabu list is a key feature of the TS, which is a deterministic method and allows movements to worse neighboring solutions if there is no improving solution. It is probable then that the TS algorithm might fall into an iterative cycle without making any progress toward an optimal solution. Cycling defines a situation whereby the TS algorithm is trapped into repetitive movements. When cycling occurs there must be intervention to break it up. The tabu list is used to prevent cycling.

Tabus can be defined and memorized in different ways. One possibility is recording a solution that is visited. Moving to a previously visited solution is forbidden. Memorizing the visited solution is the most efficient method to prevent cycling, yet it is seldom used because this requires a lot of storage making it expensive to evaluate whether or not a potential move is tabu. On the other hand, another commonly used tabu is memorizing the transformation performed on the current solution and prohibiting reverse transformations. In this manner following a move from solution X to X' precludes returning from X' to X, but arriving at X through another route is allowed.

It is recommended that transformation and reverse transformation be recorded as tabu. Recalling Figure 6.2, if the current searching point is (3,2) and the TS algorithm moves to point (2,2) as new searching point, moving back from (2,2) to (3,2) is forbidden and considered as a tabu. The TS algorithm cannot follow a previously traveled path. Yet, it is possible to return to point (3,2). Imagine that after two iterations, the search reaches point (3,3). The algorithm may move back to point (3,2) through another route that is not a tabu. This would introduce cycling because the algorithm will again move from (3,2) to (2,2), which can be solved by recording the forward transformation and the reverse transformation as tabu. In the previously mentioned example, the algorithm moves from the current searching point (3,2) to point (2,2) requires recording the moves (3,2) to (2,2) and (2,2) to (3,2) as tabu. This would eliminate cycling involving these moves.

6.6 Updating the Tabu List

Tabus are memorized for a predefined number of iterations and then they are removed from the tabu list. Generally, the basic purpose of a tabu list is to avoid returning to a previously visited point (solution). The probability of cycling caused by following a sequence of moves that ends with a previously

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visited solution is inversely related to the distance of the current searching point from that previous solution (Glover, 1989).

The number of iterations during which tabu lists are saved is called the tabu tenure (δ), which is a user-defined parameter of the algorithm. In this manner the tabu list keeps information on the last δ moves (iterations) that have been performed during the search process. For example, a solution that becomes a tabu at iteration 1 with the tabu tenure equal to 3 ($\delta = 3$) is deleted from the tabu list at iteration 4. When a new solution (information) is added to a full tabu list, it results in the removal of the oldest solution or information from the list.

Standard tabu lists usually have a fixed length. Yet, tabu lists of variable lengths have been implemented (Glover, 1989, 1990; Skorin-Kapov, 1990; Taillard, 1990, 1991).

6.7 Attributive Memory

Many types of memories for TS have been developed during past years to enhance its capacity. One of the most common of such memories is attributive memory. This type of memory records information about the properties or attributes of solutions. The most prevalent attributive memory approaches are frequency-based memory and recency-based memory.

6.7.1 Frequency-Based Memory

Frequency-based memory keeps information that facilitates selecting feature moves. Specifically, the TS algorithm saves the number of times that each solution has been selected to be the searching point, and this information is called frequency-based memory, which excises a penalty that is proportional to the frequency with which a solution is visited by subtracting a penalty value from the fitness value of the solution as follows:

$$F'(X) = F(X) - \mu \tag{6.6}$$

in which $F'(X)$ = penalized fitness function of the solution X, $F(X)$ = fitness function of the solution X, and μ = number of times that solution X has been visited. Equation (6.6) applies to a maximization problem.

The desirability of a solution is evaluated by its penalized fitness value rather than its fitness function value. Thus, frequency-based memory would select between two neighboring solutions that have the same fitness value and with the lower frequency of visitation (under maximization).

6.7.2 Recency-Based Memory

Two possible methods to record tabus were cited earlier, one that saves a solution that is visited and the other that saves the search movements that are performed.

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There is another way of recording tabus whereby attributes of solutions are memorized as a tabu list. This type of tabu list is called recency-based memory, the most prevalent memory structure applied in TS implementations. This memory structure keeps track of solutions' attributes that have changed during the recent past. For example, recalling Figure 6.2, suppose that the search starts from point (3,2) and moves to point (2,2). It is clear that in moving from (3,2) to (2,2), the value of decision variable x_1 is reduced, and by moving back from (2,2) to (3,2), increasing the value of x_1 is necessary. Moving back to the previous solution and cycling does not occur if increasing the value of x_1 is forbidden. Increasing the value of x_1 is therefore considered as a tabu for the δ next iterations. The selected attributes or properties of solutions recently visited are labeled tabu active in recency-based memory. Solutions that have tabu-active elements, or particular combinations of these attributes, are those that become tabu.

A tabu list that is constructed on recency-based memory is easy to save and to read information from. Selected attributes occur in solutions recently visited. If these attributes are labeled tabu active, then solutions that have never been visited but share the same tabu-active attributes are prevented from being visited by the search algorithm. To illustrate, suppose the search starts with point (3,2) in Figure 6.3 and moves to (2,2). Assume that δ (tabu tenure) is equal to 2. This movement decreases the value of x_1, which precludes increasing the value of x_1 in the next two iterations. The TS algorithm can thus move to (2,1), (2,3), or (1,2), but it cannot move to (3,2). Suppose that point (2,3) is selected to be the next searching point. Decreasing the value of x_2 would be a tabu in the next movement. In the third iteration the searching point is (2,3) and the two neighboring points (2,4) and (1,3) are non-tabu points, whereas the two neighboring points (2,2) and (3,3) are tabu (see Figure 6.3). It is evident in Figure 6.3 that the point (3,3) has not been visited during the search, but it is considered as a tabu because increasing the value of x_1 from 2 to 3 is tabu. Thus, it is necessary to cancel tabus in some instances. The action by which the algorithm ignores tabus is called aspiration criteria.

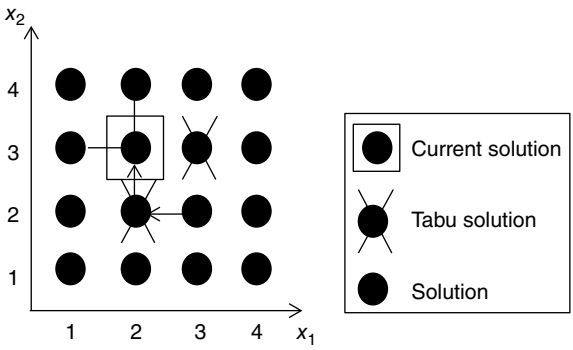


Figure 6.3 Illustration of steps based on recency-based memory in a two-dimensional decision space.

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6.8 Aspiration Criteria

Tabus may prohibit rewarding moves even when there is no danger of cycling, or they may lead to an overall stagnation of the searching process. Consequently, the aspiration criterion frees the algorithm to move to a tabu point under certain favorable conditions. A commonly used aspiration criterion is a condition in which a tabu movement leads to a better solution than the best solution that has been encountered in previous moves.

6.9 Intensification and Diversification Strategies

Intensification and diversification strategies are applied in some versions of TS. Intensification strategies have been commonly implemented based on long-term memory so that a set of elite solutions are selected, their components serve to construct new neighbors, and diversification strategies encourage examining unvisited regions by generating solutions significantly different from those searched earlier (Glover and Laguna, 1997).

These strategies ensure that all areas of the search space are adequately explored. A frequency-based tabu list keeps track of the search area. The frequency index of a previously visited solution is increased whenever it is revisited. The diversification strategy set a threshold of the frequency index. A value of the frequency index larger than a predefined threshold implies that a region has been explored frequently and the search process is restarted with a new, randomly generated point. This diversification strategy is similar to the restart mechanism of other stochastic optimization approaches. The restart in the TS, however, is guided by historical information based on an intensification strategy.

6.10 Termination Criteria

The termination criterion determines when to terminate the TS algorithm. Selecting a good termination criterion has an important role for the correct convergence of the algorithm. The number of algorithmic iterations, the amount of improvement of the objective function between consecutive iterations, and the run time are common termination criteria for the TS algorithm.

6.11 User-Defined Parameters of the TS

The value of the tabu tenure (δ) and the criterion used to decide when to terminate the algorithm are the user-defined parameters of the TS. Choosing an appropriate δ is important. Large values of δ make the algorithm to move

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gradually away from points visited during the previous δ iterations. A small value for δ provides more mobility to the algorithm. Previous experience with the TS reveals that there is a stable range of δ that prevents cycling and leads to good solutions. A good choice of the parameters is related to the decision space of a particular problem. In general, optimal parameters for one problem may not function well for other problems. Consequently, determining a good parameter setting often requires the execution of a large number of time-consuming experiments. The setting of suitable parameters relies principally on in-depth experience with the type of problem being solved. Yet, a reasonable method for finding appropriate values of the TS parameters is performing sensitivity analysis whereby combinations of parameters are tested with multiple runs for each combination. From these results the analyst may gain clear guidance on the choice of parameters that produce near-optimal solutions.

6.12 Pseudocode of the TS

```
Begin
  Input parameters of the algorithm and initial data
  Let X*=the best point and X=the current search
    point
  Generate a search point (X) randomly and evaluate
    its fitness value
  Set X*=X
  While (termination criteria are not satisfied)
     Generate neighbor points around the searching
       point and evaluate their fitness values
     If all neighbor points are tabus and cannot satisfy
       the aspiration criteria
          Stop the algorithm and report the best point (X*)
     End if
     Select the best neighbor point which is not tabu
       or satisfies the aspiration criteria
     Put X=the selected point
     If X is better than X*
         Set X*=X
     End if
     Update the tabu list
  End while
  Report the best point (X*)
End
```

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6.13 Conclusion

This chapter described the TS, which is a meta-heuristic algorithm for combinatorial optimization. First, a brief literature review of development and applications of the TS was presented. This was followed by a description of the fundamentals of the TS and its algorithmic nuances. A pseudocode of the TS closed this chapter.

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